Global Reporting Regulation, Stock Liquidity, and Policy Diffusion

Abstract

We examine stock liquidity around the implementation of 17 families of reporting regulation and study their international diffusion. The issuer outcome panel contains 1,055,142 security-years in 90 jurisdictions. Separate family-specific cohort comparisons yield a first-year log-spread estimate of -0.049 with equal family weights and +0.015 with market-capitalisation family weights. These estimates describe geographic policy exposure. A separate source-corroborated issuer analysis covers auditor reporting, digital filing, conflict minerals and banking disclosure. Across these four selected families, first-year log-spread estimates are -0.022 and +0.209 under the two weighting schemes. Longer-window support is narrower, and pretrends limit causal interpretation. The diffusion models provide no evidence that survives broad multiplicity adjustment. The findings show why legal scope, information timing and economic weights matter when combining heterogeneous reporting policies.

1. Introduction

Disclosure regulation can improve liquidity by supplying credible information to investors. Whether this channel operates depends on what is disclosed, who receives it, and which firms are required to report. A confidential tax return, a public auditor report and a crypto-asset white paper do not impose the same information shock on ordinary equity markets. We ask how liquidity changes around these rules and whether prior foreign implementation helps explain subsequent policy choices.

We follow the empirical logic of Krueger et al. (2023): establish the reporting obligation, connect it to firms and information timing, examine complementary liquidity outcomes, and assess identification before drawing conclusions. Our setting extends beyond comprehensive sustainability reporting to 17 economic regulation families. We then use the competition, learning, emulation and coercion framework in Koning, Mertens and Roosenboom (2018) to examine diffusion.

The analysis distinguishes two populations. The broad panel assigns geographic exposure to individual vendor-issuer outcomes on country-family calendars. The narrower studies require source evidence about covered issuers or actual reports. This distinction permits a broad comparison without implying that every company in a country is legally covered. Banking phases enter each aggregate once.

Table 2 gives equal and market-capitalisation family weights equal prominence. The first-year equal-family exposure estimates indicate lower spreads and price impact; the market-capitalisation estimates are less favourable. Source-corroborated issuer results also differ by family. These differences identify the population and policy mix to which an estimate applies. They do not establish a common causal benefit from reporting regulation.

2. Institutional setting and research design

Public reporting can reduce adverse selection when it supplies incremental information. Enforcement, prior disclosure and compliance costs can moderate the effect. Comprehensive mandatory E/S/G requires a response on environmental, social and governance topics; the final topic determines the joint date. Comply-or-explain requires a response, while CSR spending and climate-only rules do not establish all-topic disclosure. The inherited review retains 37 qualifying country phases from 41 reviews; earliest separate topic histories remain incomplete. In the reference study, voluntary earnings guidance is an information-environment proxy, rather than a voluntary ESG treatment.

Eligibility is rule specific (Table 1). Confidential CbCR goes to tax authorities; its actual return date is not a public disclosure date. Public bank CbCR attaches to credit institutions and consolidation perimeters, with partial and full phases. Listed holding companies are not automatically the regulated reporter. Beneficial-ownership rules can exempt securities-reporting issuers, while MiCA concerns crypto assets and services. Country calendars therefore cannot certify ordinary-equity coverage.

The 2000-2025 market panel contains 99,100 vendor quote/company identifiers and 1,055,142 security-years. Outcomes are the log median relative quoted spread, log median Amihud price impact and the fraction of valid observed returns equal to zero. Each outcome requires 120 valid observations; logged proxies use annual 1/99 winsorisation. The calendar-year and observed-return denominator differ from the reference paper. A vendor company code is not a complete historical legal-entity register.

For each family, cohort and endpoint, we subtract the weighted change from year g-1 among known later implementers from the change among implementers in g. We select one company home quote by pre-event market capitalisation and require fixed issuer, country and outcome coverage. Geographic studies require at least 30 issuers per country and three comparison markets. First-year contrasts use g-1 and g; balanced contrasts fix support from g-3 to g+2 and average years 0-2. Unknown histories are never classified as never-treated.

GDP in g-1 is allocated across sampled issuers within countries. In covered studies, the denominator is the fixed source-study country universe, so each cohort does not receive a fresh country GDP mass. Cohort weights equal treated economic mass. Across families, equal weights assign 1/F; market-cap weights use summed pre-event treated issuer capitalisation. Equal-issuer and issuer-cap within-family alternatives are disclosed. Repeated policy exposures are not a unique-issuer investment portfolio.

Country influence functions preserve covariance across geographic families; 1,999 shared multiplier draws yield conditional intervals, lead diagnostics and Holm adjustment across 12 principal coefficients. Collective legal waves, concurrent reforms and spillovers still limit policy inference. Covered-study intervals describe issuer sampling only. Few common regulatory waves do not support causal policy p-values. Their minimum of three treated and twenty comparison issuers permits sparse case studies, not a representative global census.

3. Liquidity results

Table 2 Panel A reports issuer outcomes under geographic exposure. The first-year contrast supports 15 families. Equal-family log spread is -0.049, approximately a 4.8% decline, log price impact is -0.072, and the zero-return fraction falls by 0.00764, or 0.764 percentage points. With market-cap family weights, the corresponding estimates are +0.015, +0.009 and -0.00249. Their conditional intervals include zero. Confidential tax CbCR receives 33.1% of market-cap exposure weight; that audience is tax authorities rather than public investors.

The balanced comparison supports nine families. Equal-family log spread is -0.032 and log price impact is -0.013; market-cap values are +0.040 and +0.095. The family set changes with the horizon, so these contrasts are not a dynamic continuation of the 15-family estimate. Figure 1 reports every attempted family. Figure 2 shows balanced profiles; joint lead diagnostics reject for spreads under both schemes. The complete estimates include timing-shift and independent-choice sensitivities.

Table 3 extends covered-issuer evidence beyond banking. US CAM eligibility uses reported filer class, EGC status, fiscal end and the PCAOB phase; SEC acceptance dates determine filing years. First supported required-report cohorts contain 114, 1,025 and 625 issuers in 2019-2021 before complete-outcome screens. The estimated first-year sample contains 1,481 treated issuer-cohort observations and 2,767 comparison issuer-cohort observations. Comparisons require known nonrequirement in every post year, not an absent checkbox. CAM log spread is +0.236 and price impact is +0.207.

The early US XBRL study uses explicit mandatory-compliance cover checks and filer class, rather than unchecked boxes. Six treated issuers and 62 later-phase comparisons have complete first-year outcomes; log spread is -0.329. The comparisons are valid only through 2009 and cannot support years 0-2. Apple, Intel and Cisco supply three first-cycle conflict-minerals reporters. Their first-year log spread is -0.085, but price impact and zero returns rise. The REIT comparison screen does not certify absence of all incidental manufacturing activity.

The observed banking publication study has three matched full-report publishers and 21 first-year comparisons; log spread is +0.089. Actual dates are 20 March 2015 for Deutsche Bank, 27 March for Aareal and 30 June for HSBC. HSBC is a listed parent reporter. Publication does not necessarily mark the first-ever voluntary disclosure. A separate statutory initial-bank sensitivity retains the larger institution sample; the two banking phases never enter the same aggregate.

Panel B of Table 2 combines the four supported observed-report/phase studies. Equal-family first-year log spread is -0.022; market-cap log spread is +0.209, dominated by the CAM sample. Price impact and zero-return fractions increase under both schemes. Balanced support contains only banking and conflict minerals. These sparse, source-selected populations and different industry exposures prevent generalisation to all 17 legally covered issuer populations.

4. International diffusion and conclusion

Following Koning et al., competition uses prior implementation among export-pattern peers; learning uses prior outcomes among geographic neighbours; emulation uses language, legal-origin and IGO peers; external dependence uses IMF credit and aid. Signals precede the focal decision. Family-year strata and country-clustered covariance accompany timing-logit and status alternatives, with lagged income, growth and trade controls where specified. The ledger identifies first recorded phases, not certified first-ever adoptions.

Across 1,330 attempted designs, 433 stable focal coefficients remain; none passes the broad Holm adjustment. The issuer filing evidence creates no new independent national adoption choices, so the entire matrix is refitted to the unchanged lagged policy input. We do not insert later liquidity estimates into earlier learning signals.

Reporting rules differ in their covered firms, audience and information dates. Liquidity associations change with both family weights and issuer scope. The evidence motivates a conditional account of disclosure effects, while pretrends and incomplete eligibility histories constrain causal conclusions. Complete source dispositions, unsupported cells and alternative specifications accompany the exhibits.

Table 1. Implemented regulation families and issuer evidence

Economic regulation family	Geographic support First / balanced	Covered-issuer evidence / remaining limit
Audit oversight / rotation	Yes / Yes	Auditor/client history needed
Expanded auditor reporting	Yes / Yes	US CAM filing cohorts
Beneficial ownership	Yes / Yes	Listed exemptions; venue gaps
Banking public CbCR	Yes / Yes	Bank perimeter; 3 dated publishers
Confidential tax CbCR	Yes / Yes	Nonpublic returns; UPE gaps
General public tax CbCR	Yes / No	No observed eligible filing cohort
Climate disclosure	Yes / Yes	Historical thresholds/report gaps
Sectoral climate reporting	No / No	Sector and panel boundary
Digital assets	Yes / Yes	Crypto object; equity scope differs
Comprehensive mandatory E/S/G	Yes / Yes	37 country phases; issuer gaps
Extractive payments	Yes / No	Sector/Form SD report gaps
Corporate governance	Yes / No	Listing and size phases missing
IFRS reporting	Yes / No	Prior GAAP and report gaps
Supply-chain reporting	Yes / Yes	3 US first-cycle reporters
Financial sustainability disclosure	Yes / No	Late phases; actual reports missing
Informed sustainability reporting	No / No	Scope and calendar unsupported
XBRL / digital filing	Yes / No	US early phase; first exhibit unknown

Coverage through 2025 only; future schedules are excluded. Geographic support is outcome support, not verified legal eligibility. Covered issuer studies support four families in the first year and two in the balanced observed-report subset. Country and issuer thresholds, exemptions, reporting periods and actual publication dates remain separate. Full legal scopes and event-level gap overlays are online.

Table 2. Both cross-family weighting schemes

Population / coverage / outcome	Equal family Estimate [interval]	Market-cap family Estimate [interval]
A. Geographic issuer exposure First year (15 families): Log spread	-0.049 [-0.082, -0.016]	+0.015 [-0.027, +0.056]
A. Geographic issuer exposure First year (15 families): Log price impact	-0.072 [-0.115, -0.030]	+0.009 [-0.059, +0.078]
A. Geographic issuer exposure First year (15 families): Zero-return fraction	-0.0076 [-0.0105, -0.0048]	-0.0025 [-0.0063, +0.0014]
A. Geographic issuer exposure Years 0-2 (9 families): Log spread	-0.032 [-0.088, +0.023]	+0.040 [-0.039, +0.120]
A. Geographic issuer exposure Years 0-2 (9 families): Log price impact	-0.013 [-0.080, +0.054]	+0.095 [-0.025, +0.215]
A. Geographic issuer exposure Years 0-2 (9 families): Zero-return fraction	-0.0018 [-0.0077, +0.0040]	+0.0067 [-0.0073, +0.0206]
B. Source-corroborated subset First year (4 families): Log spread	-0.022 [-0.098, +0.053]	+0.209 [+0.170, +0.248]
B. Source-corroborated subset First year (4 families): Log price impact	+0.029 [-0.130, +0.188]	+0.201 [+0.135, +0.267]
B. Source-corroborated subset First year (4 families): Zero-return fraction	+0.0071 [+0.0018, +0.0125]	+0.0065 [+0.0042, +0.0088]
B. Source-corroborated subset Years 0-2 (2 families): Log spread	+0.023 [-0.126, +0.173]	-0.034 [-0.143, +0.074]
B. Source-corroborated subset Years 0-2 (2 families): Log price impact	+0.293 [+0.058, +0.528]	+0.339 [+0.185, +0.493]
B. Source-corroborated subset Years 0-2 (2 families): Zero-return fraction	+0.0112 [+0.0032, +0.0192]	+0.0083 [+0.0029, +0.0137]

GDP weights within families. Panel A intervals condition on country sampling and the observed policy calendar; Panel B intervals describe issuer sampling only and are not causal policy confidence intervals. Zero returns are fractions. Samples are fixed within each horizon; family composition differs across horizons and populations. Equal-issuer/cap within-family alternatives and full diagnostics are online. No unsupported family is assigned a zero estimate.

Table 3. Covered-issuer first-year comparisons beyond banking

Source-corroborated study	Treated / comparison issuer-cohort observations	Log spread	Log price impact	Zero-return fraction
US auditor CAM (2019-21)	1,481 / 2,767	+0.236	+0.207	+0.0063
US early XBRL (2009)	6 / 62	-0.329	-0.417	-0.0004
US conflict minerals (2014)	3 / 70	-0.085	+0.249	+0.0071
Observed bank full reports (2015)	3 / 21	+0.089	+0.077	+0.0155
Alternative: bank initial phase (2014)	23 / 21	+0.035	-0.343	+0.0006

Counts are issuer-cohort observations; comparison issuers can recur across CAM cohorts. GDP is allocated over the fixed study country universe. The observed-report aggregate uses the first four rows. The last row replaces banking full reports in a separate sensitivity. Sparse, selected treated groups, industry differences, common waves and subsidiary spillovers limit interpretation; REIT comparisons are industry screened rather than legally certified noncovered. Full estimates include balanced and alternative within-family weights.

Source review and timing

Explicit cover and font evidence resolves EGC status for 9,382 of 13,122 linked filings and primary filer status for 11,278; 20 conflicting declarations remain withheld. All 30 sampled CAM auditor sections are confirmed. The XBRL review inspects 1,385 filings. Apple and Intel have verified actual filing days; Cisco retains year precision. Irish crosslisted examples show that trading location differs from legal incorporation, but actual first publication dates remain unresolved. No missing report is coded as voluntary or exempt merely because it is absent.

References and primary sources

Krueger, P., Z. Sautner, D. Y. Tang and R. Zhong. 2023. The Effects of Mandatory ESG Disclosure Around the World. Supplied January/February 2023 working paper; published in Journal of Accounting Research (2024), doi:10.1111/1475-679X.12548.

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SEC. Interactive Data Compliance and Disclosure Interpretations, Q105.04; Release 33-9002. sec.gov/files/divisions/corpfin/guidance/interactivedatainterp.htm.

SEC. Conflict Minerals, Release 34-67716 (2012). Federal Register official copy: public-inspection.federalregister.gov/2012-21153.pdf. Apple, Intel and Cisco first-cycle Form SD sources and date precision appear in the online evidence register.

IRS. T.D. 9773, Internal Revenue Bulletin 2016-29. irs.gov/irb/2016-29_IRB. Bank report sources, Irish statutory accounts, legal scope sources and all attempted downloads are recorded in the online source register.

Figure 1. Individual issuer outcomes across regulation exposures

First-year geographic exposure contrasts. All 17 attempted families appear; 15 have common outcome support. Bounds condition on country sampling and are not a certificate of eligible-issuer coverage. Right-hand weights use sampled treated capitalisation in the pre-event year.

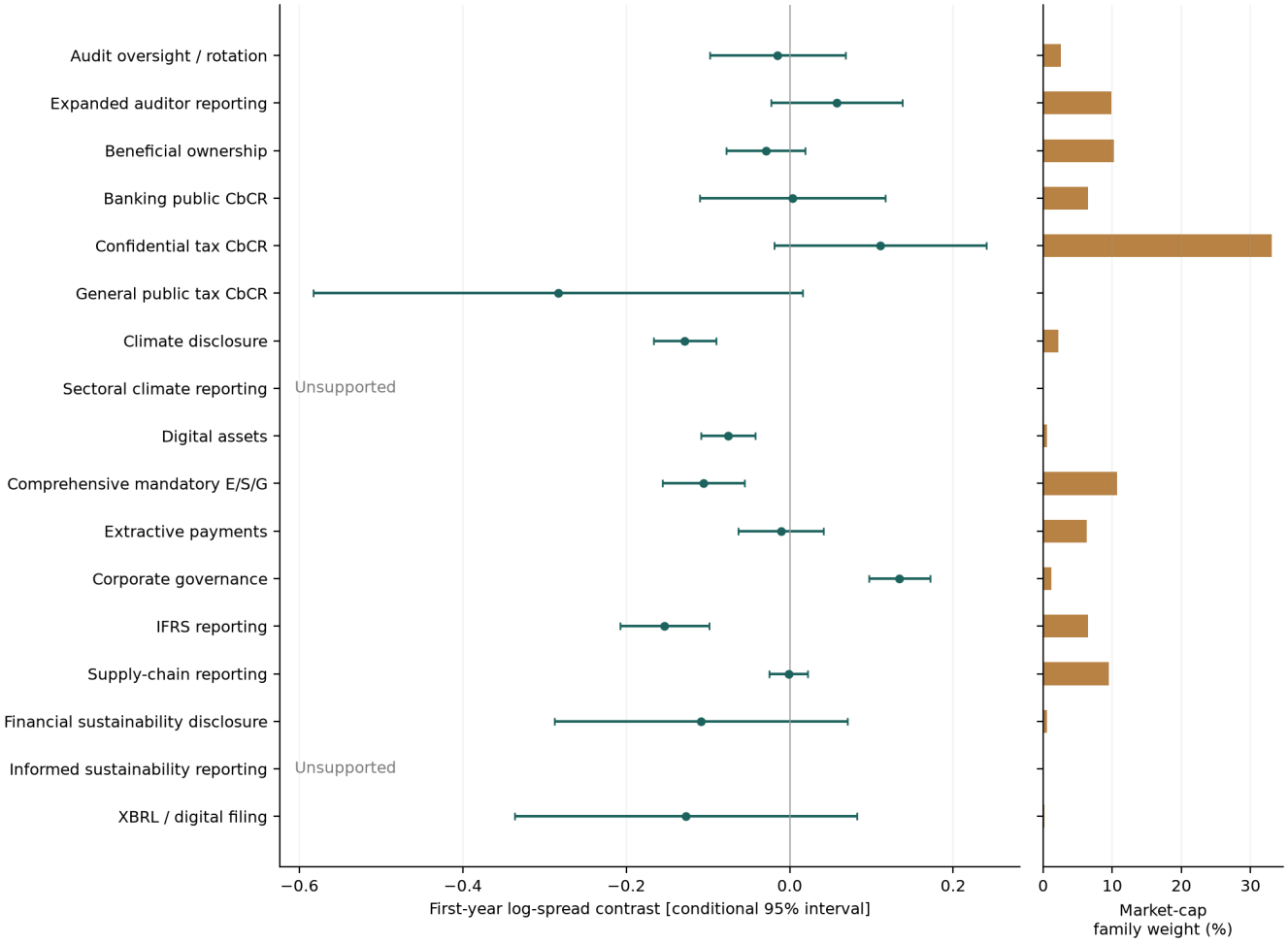
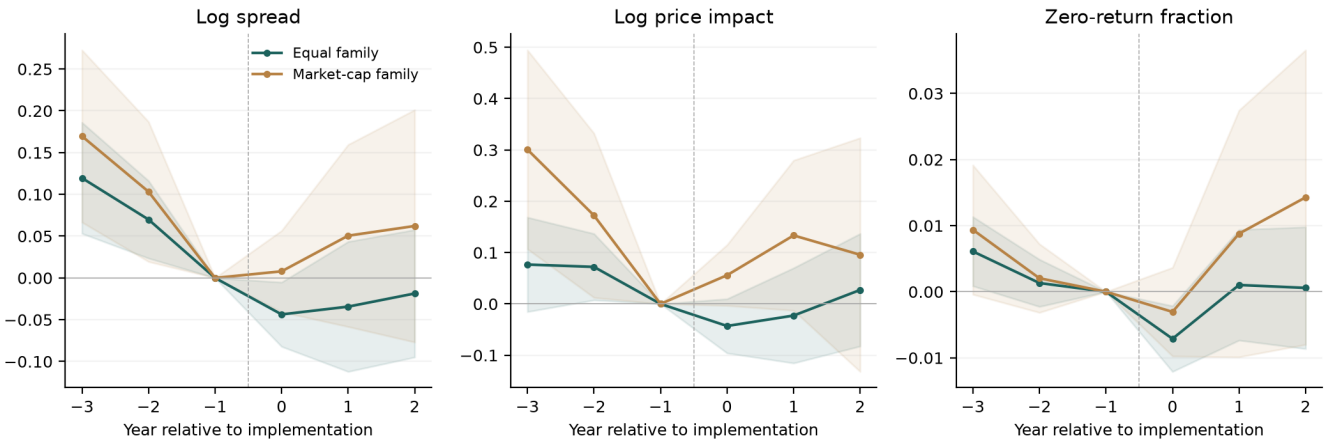


Figure 2. Balanced issuer exposure profiles under both weighting schemes

Nine common-support families, fixed issuers and cohorts from g-3 to g+2. Year -1 is the zero reference. Shading gives simultaneous country-sampling bands. Joint lead diagnostics reject for log spreads under both schemes; profiles do not establish parallel trends or causal policy effects.



Reading the exhibits

The broad exposure panel and the source-corroborated studies answer different questions. The first tracks issuer outcomes around geographic regulation calendars; the second conditions on specific legal or report evidence. A favourable geographic coefficient cannot close an eligibility gap. First-year and balanced windows also change support, and should not be joined into one event path.

Reproducible evidence

The companion report provides the frozen design, all family and aggregate estimates, weight contributions, coverage failures, alternative weights, source dispositions and 441 additive event-review JSONs. Licensed issuer and daily market data remain local. The complete diffusion matrix includes failures and coefficients opposing the hypothesised mechanisms. https://hunternbh.github.io/policy-diffusion/